

PROGRAM – 9

Organize Kubernetes Pods Using Labels and Namespaces

Project Name: program_devops_9

Technologies: Kubernetes, Minikube, YAML

Step 1: Ensure Minikube is running

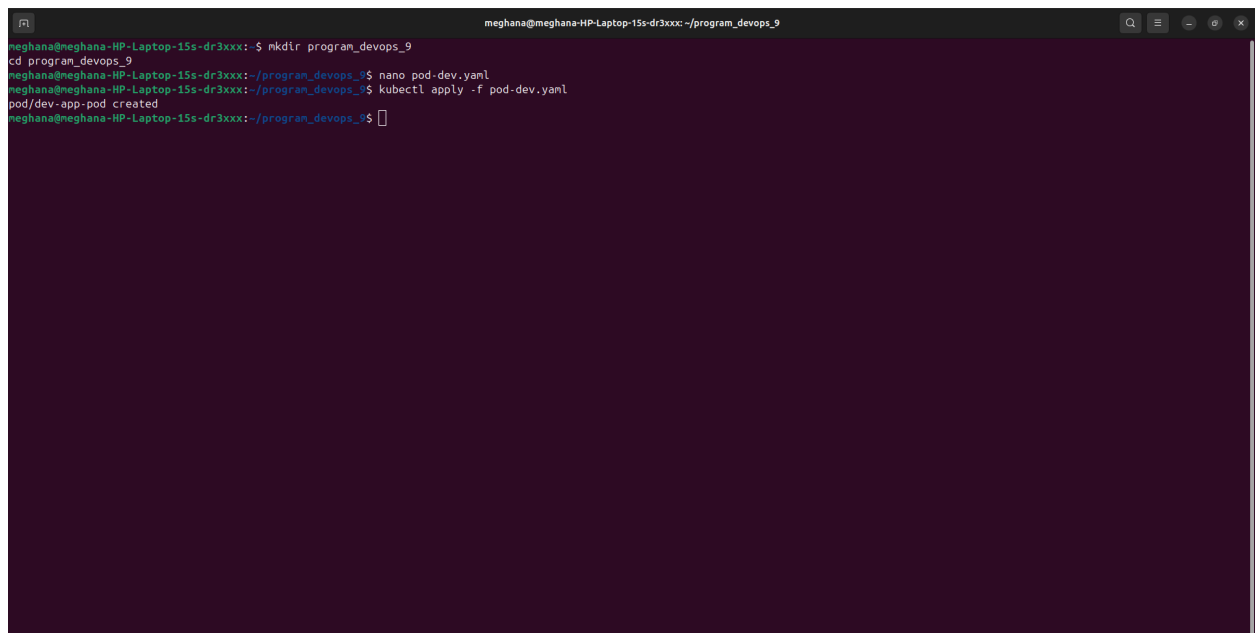
minikube start

Step 2: Verify cluster

kubectl get nodes

Step 3: Create a new Namespace

kubectl create namespace dev

A terminal window with a dark purple background and white text. The window title is 'meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_9'. The terminal shows the following commands and output:

```
meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program_devops_9
meghana@meghana-HP-Laptop-15s-dr3xxx: $ cd program_devops_9
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ nano pod-dev.yaml
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl apply -f pod-dev.yaml
pod/dev-app-pod created
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$
```

Step 5: Create project directory

mkdir program_devops_9

cd program_devops_9

Step 6: Create Pod YAML with labels

nano pod-dev.yaml

```
GNU nano 7.2 pod-dev.yaml
apiVersion: v1
kind: Pod
metadata:
  name: dev-app-pod
  namespace: dev
  labels:
    app: web
    env: dev
spec:
  containers:
  - name: nginx-container
    image: nginx
    ports:
    - containerPort: 80
```

Step 7: Create Pod in namespace

kubectl apply -f pod-dev.yaml

Step 8: Verify Pod in dev namespace

kubectl get pods -n dev

Step 9: View Pod labels

kubectl get pods -n dev --show-labels

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl get pods -n dev
NAME                READY   STATUS    RESTARTS   AGE
dev-app-pod         1/1     Running   0           25s
python-dev-pod      0/1     CrashLoopBackOff   61 (2m56s ago)    18d
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl get pods -n dev --show-labels
NAME                READY   STATUS    RESTARTS   AGE   LABELS
dev-app-pod         1/1     Running   0           29s   app=web,env=dev
python-dev-pod      0/1     CrashLoopBackOff   61 (3m ago)    18d   app=python,env=dev
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl get pods -n dev -l app=web
NAME                READY   STATUS    RESTARTS   AGE
dev-app-pod         1/1     Running   0           32s
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl describe pod dev-app-pod -n dev
Name:               dev-app-pod
Namespace:          dev
Priority:             0
Service Account:     default
Node:               minikube/192.168.49.2
Start Time:          Mon, 05 Jan 2026 13:11:12 +0530
Labels:              app=web
                    env=dev
Annotations:          <none>
Status:              Running
IP:                  10.244.0.94
IPs:                 IP: 10.244.0.94
```

Step 10: Filter Pods using labels

```
kubectl get pods -n dev -l app=web
```

Step 11: Describe Pod (namespace-specific)

```
kubectl describe pod dev-app-pod -n dev
```

Step 12: Create another namespace (optional)

```
kubectl create namespace prod
```

```
kubectl apply -f pod-prod.yaml
```

Step 14: Verify Pods across namespaces

```
kubectl get pods --all-namespaces
```

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ nano pod-prod.yaml
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl apply -f pod-prod.yaml
pod/prod-app-pod created
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl get pods --all-namespaces
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
default	demo-app-787c9b7c57-nmffk	1/1	Running	6 (2m49s ago)	18d
default	demo-app-787c9b7c57-q7bz4	1/1	Running	6 (2m49s ago)	18d
default	my-first-pod	1/1	Running	2 (2m49s ago)	170m
default	nginx-deployment-599786d4bc-6xdvs	1/1	Running	4 (2m48s ago)	4d22h
default	nginx-deployment-599786d4bc-chbsf	1/1	Running	4 (2m49s ago)	4d22h
default	nginx-pod	0/1	Completed	0	172m
default	nodejs-pod	1/1	Running	6 (2m39s ago)	18d
default	program-devops-7-deployment-56bf5757cf-2qqqg	1/1	Running	1 (2m48s ago)	26m
default	program-devops-7-deployment-56bf5757cf-p9cqg	1/1	Running	1 (2m48s ago)	26m
default	program-devops-8-pod	1/1	Running	1 (2m39s ago)	17m
default	python-hello-7f4448c876-zqdz7	1/1	Running	8 (2m39s ago)	18d
dev	dev-app-pod	1/1	Running	0	64s
dev	python-dev-pod	0/1	CrashLoopBackOff	61 (3m35s ago)	18d
kube-system	coredns-66bc5c9577-fn42b	1/1	Running	12 (2m44s ago)	40d
kube-system	etcd-minikube	1/1	Running	12 (2m41s ago)	40d
kube-system	kube-apiserver-minikube	1/1	Running	12 (2m39s ago)	40d
kube-system	kube-controller-manager-minikube	1/1	Running	13 (2m48s ago)	40d
kube-system	kube-proxy-zpvt8	1/1	Running	12 (2m49s ago)	40d
kube-system	kube-scheduler-minikube	1/1	Running	12 (2m49s ago)	40d
kube-system	storage-provisioner	1/1	Running	24 (2m29s ago)	40d
kubernetes-dashboard	dashboard-metrics-scraper-77bf4d6c4c-c7kns	1/1	Running	4 (2m49s ago)	4d22h
kubernetes-dashboard	kubernetes-dashboard-855c9754f9-jl9lh	1/1	Running	7 (2m29s ago)	4d22h
prod	prod-app-pod	0/1	ContainerCreating	0	3s

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$
```

Step 15: Delete namespace (cleanup)

```
kubectl delete namespace dev
```

```
kubectl delete namespace prod
```

```
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_9$ kubectl delete namespace dev
kubectl delete namespace prod
namespace "dev" deleted
```

